post_holes_dug = 4
posts_set = 4
roof_panel_ft2_set = 216.01
wall_panel_ft2_set = 366.02
flows:
dig-post-holes: ran 4 times; moved budget_usd -129.3, cost_usd 129.3, post_holes_dug 4
set-posts: ran 4 times; moved posts_set 4
set-floor-panels: ran 1 times; moved budget_usd -584.77, cost_usd 584.77, floor_panel_ft2_set 63.98
set-wall-panels: ran 8 times; moved budget_usd -3345.46, cost_usd 3345.46, wall_panel_ft2_set 366.02
set-roof-panels: ran 1 times; moved budget_usd -2363.13, cost_usd 2363.13, roof_panel_ft2_set 216.01
cut-openings: ran 3 times; moved openings_cut 3
set-windows-and-door: ran 3 times; moved budget_usd -1972.78, cost_usd 1972.78, openings_set 3
frame-loft: ran 7 times; moved budget_usd -45.82, cost_usd 45.82, joists_set 7
lay-subfloor: ran 1 times; moved budget_usd -98.66, cost_usd 98.66, deck_sheets_laid 2
install-boiler: ran 1 times; moved boiler_installed 1, budget_usd -5947.5, cost_usd 5947.5

ASSUMPTIONS, EVERY SETTING WHOSE REASON SAYS IT IS ASSUMED

weather.inside_celsius = 20 celsius; reason: assumed design day
weather.outside_celsius = 0 celsius; reason: assumed design day
weather.sun_watt_per_square_metre = 500 w_per_m2; reason: assumed clear-sky peak; the solver applies it to every side
finish.mean_reflectance = 0.5 ratio; reason: assumed mid-grey interior; Jim's wall finish gives it
finish.sky_angle_degrees = 65 degree; reason: assumed open sky; the site gives it
doors.offset_mm = 0 mm; reason: assumed: the west end of the north wall; which end the door is at is open
joists.spacing_mm = 406 mm; reason: assumed: 16 in on centre
ledgers.member_width_mm = not given mm; reason: assumed: a ledger cut from the joist stock, fastened to the wall's inside face; the fastening is not checked
ledgers.member_depth_mm = not given mm; reason: assumed: a ledger cut from the joist stock, as deep as the joists it carries
header.plies = 2 count; reason: assumed: two plies side by side, 3-1/2 in of LVL or 3 in of 2x lumber, fit inside the SIP's 3-5/8 in core (sip.prs610_core_thickness)
roof_loads.roof_dead_load_psf = 15 psf; reason: assumed: 15 psf, the roof dead load Boise's roof-only header look-up assumes, research row gl.hdr.vl_lookup_roof_only; the roof build-up gives it
roof_loads.deflection_limit_ratio = 240 ratio; reason: assumed: span over 240, the live-load deflection limit of Boise's header tables, research row gl.hdr.vl_limits, applied to the whole load
roof_loads.roof_bears_on = south, north compass_sides; reason: assumed: the roof's 8 ft south hangover can only be carried by members running north-south across the south wall, so the roof bears on the south and north walls

UNMEASURED PIECES

none

PROJECT	SHEET	SHEET NUMBER	DATE
Small house on posts, from Jim's pencil sketch	**Design summary**	S-1 (page 3 of 3)	2026-10-20
CONFIG COMMIT	ENGINE PIN	SCALE	SOURCE
31254d4	we 5af4c5f; rendered with we c3542b1	not to scale	we geometry